

Arthur Domingues Fachel Nunes

Data Analyst

Porto Alegre, Brazil · [LinkedIn](#) · [GitHub](#) · arthur.nunes6889@gmail.com

EDUCATION

Pontifical Catholic University of Rio Grande do Sul

Jan 2024 – Dec 2029

Scientific Initiation Scholarship (IC)

- Academic presentation evaluated 4.5 / 5.0 at PUCRS Academic Week 2025.
- Active member of MALTA-LAB (Machine Learning and Theory Lab) since Aug 2024.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Aug 2024 – Present

MALTA-LAB (Machine Learning Theory Lab and Applications) — PUCRS

- Co-authored three scientific papers across two research projects; one accepted at BRACIS 2026.
- Designed, implemented, and evaluated RAG architectures for domain-specific geoscience question answering.
- Conducted data analysis and visualisation with Matplotlib and PyTorch for stratigraphic datasets.

RESEARCH PROJECTS

Petrobras: Fine-tuning LLMs for Geoscience

2024 – Present

- Fine-tuned a large language model on geological corpora for downstream field tasks.
- Analysed stratigraphic data using Scikit-learn and Pandas; applied clustering algorithms for data exploration.
- Developed a Flask API wrapping the local LLM for real-time inference by geologists in the field.
- Built an automated pipeline using Hermes Agent to parse and structure geological Markdown files into vector-searchable documents for RAG, scheduled and maintained through cron-based workflows.

DESINFORMAÇÃO: Misinformation Detection via RAG

2024 – 2025

- Automated web scraping of Brazilian news outlets with Selenium for misinformation article collection.
- Created analytical benchmarks and evaluation pipelines for misinformation classification.
- Integrated scraped data with LLMs through a Retrieval-Augmented Generation (RAG) architecture.

Brazil Literacy: Municipal Data Analysis

2025 – Present

- Analysed municipal-level data linking literacy rates, basic sanitation coverage, and school infrastructure across Brazil.
- Building an interactive dashboard with dedicated API endpoints per data dimension.

PUBLICATIONS

Nunes, A. F., Parraga, O., Spigolon, A. L. D., de Barros, G., Gonzaga Jr, L., Veronez, M. R., Barros, R. C., & Kupssinskü, L. S. (2026). *The Role of Contextual Grounding in Geoscience Benchmarks*. In BRACIS 2026 — Brazilian Conference on Intelligent Systems. Accepted.

TECHNICAL SKILLS

Programming Languages	Python, SQL, C, Java, JavaScript
Frameworks & Libraries	PyTorch, Flask, Scikit-learn, Pandas, Matplotlib, Selenium, BeautifulSoup
Data & ML	Machine Learning, Deep Learning, NLP, Data Analysis, AWS Bedrock, MySQL
Web & APIs	REST APIs, Web Scraping (Selenium, BS4), AWS (Bedrock, Lightsail, EC2), Firebase
Version Control	Git, GitHub
Languages	Portuguese (Native), English (Fluent), Spanish (Advanced), Finnish (Intermediate)